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31338 Network Servers
32520 Systems Administration

Week 2

Installation & configuration

(Documentation, Hardware and kernel configuration
System installation and management)

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Documentation

- Linux: command-based OS using CLI with options
- Man pages (manual) – view with `man` command

```
man passwd
man 5 passwd #searches for passwd in section 5 of the man pages
```

- Man sections:
  - **1** = General commands (normal users)
  - **2** = System calls (programmers)
  - **3** = C library functions (programmers)
  - **4** = Special files (usually /dev devices)
  - **5** = File formats and conventions
  - **6** = Games
  - **7** = Miscellanea
  - **8** = System administration commands and daemons
  - **9** = Kernel Routines

Configuring the Linux manual

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- Config file is `/etc/man_db.conf` (or `/etc/man.config`)
  - specify the locations to search for man pages.
- Locations to search for man pages

```
MANDATORY_MANPATH /usr/man
MANDATORY_MANPATH /usr/share/man
MANDATORY_MANPATH /usr/local/share/man
```

- Sections of the manual and search order if not specified

```
SECTION 1 1p 8 2 3 3p 4 5 6 7 9 0p n l p o 1x 2x 3x 4x 5x 6x 7x 8x
```

- Package documentation
  - installed when software packages are installed (e.g. Red Hat Package Manager - rpm)
  - often in `/usr/doc` or `/usr/share/doc`
  - may be text files, HTML, PDF – depends on package

Hardware configuration

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- PC hardware: PCI bus, RS-232 serial ports, USB, sound cards, video cards, IDE/ATA and SCSI disks through Motherboard.
- PCI (PCI-e, Express)= Peripheral Component Interconnect
  - modern way of connecting peripherals to PCs
  - PnP built-in
  - **`lspci`** list PCI devices

- USB = Universal Serial Bus (1.0, 1.1, 2.0, 3.0, 3.1, 3.2, 4)
  - USB 1.1 = 12Mbps, USB 2.0 = 480Mbps, USB 3 up to 20Gbps . . . .
  - **`udev`** (config in **`/etc/udev`**; **`man udev`**) and need to install the system-dev package
    - **`udev is`** device manager for Linux kernel. It runs as a daemon on a Linux system and listens (via netlink socket) to events the kernel sends out if a new device is initialized or a device is removed from the system.

IDE/SATA/SCSI disks

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To connect a hard drive to a Linux system, the Linux kernel assigns the drive device a file in the /dev directory

- IDE = Integrated Drive Electronics
  - Very old, difficult to find nowadays
  - Device files **/dev/hda, /dev/hdb, /dev/hdc,** …
- Current PCs have serial ATA (SATA) drives, SATA 3 (3.5) (6 Gbit/s)
  - Device files **/dev/sda, /dev/sdb,** …**,/dev/sdz**
- SCSI = Small Computer System Interface
  - separate bus for peripherals, usually disks & tapes
  - many varieties: SCSI-1, SCSI-2, SCSI-3, Fast, Wide, Ultrawide
  - usually need a SCSI host adapter (PCI card) in system, BIOS
  - Linux: **cat /proc/scsi/scsi #**To list scsi devices on Linux system
  - Disk devices: **/dev/sda, /dev/sdb**, etc, based on SCSI ID 0, 1, …
- SCSI has been replaced by USB or SATA interfaces

NVMe disk

- NVME = Non-Volatile Memory Express
  - The newest, fastest protocol specifically designed for flash-based Solid-State Drives (SSDs)
  - Communicates directly with the CPU using the computer's high-speed PCIe (Peripheral Component Interconnect Express) bus.
  - Modern NVMe drives (PCIe 4.0/5.0) can reach read/write speeds ranging from 3,500 MB/s to over 7,000 MB/s.
  - Device files **`/dev/nvme0n1p1, /dev/nvme0n1p2,`** …

IDE

SATA

SCSI

NVME

Learning about the kernel & modules

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- **`uname -a`** # to retrieve essential information about the system, print system information, -a: all, -s: kernel name

  ```
  Linux localhost.localdomain 4.18.0-193.6.3.el8_2.x86_64 #1 SMP
  Wed Jun 10 11:09:32 UTC 2020 x86_64 x86_64 x86_64 GNU/Linux
  ```

  – *system nodename release version machine processor hardware OS*

- Older kernels were monolithic with all modules compiled in
- Current kernels still monolithic, but support loadable modules
  – modules loaded if/when needed

- **`lsmod`** #prints the contents of the **`/proc/modules`** file
  – It shows which loadable kernel modules are currently loaded.
- **`modinfo`** *<modulename>*
  – shows module info about a specific module, e.g. bluetooth

Loading & removing kernel modules

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- Two ways to load a kernel module into the kernel:

  `insmod /lib/modules/<version>/kernel/fs/fat/fat.ko`

  - requires exact module filename, saved in `/lib/modules`,
  - doesn't check dependencies

  `modprobe fat` "a filesystem" #to add and remove modules from Linux Kernel

  - requires module name (not filename); –r (remove) modules
  - also loads any modules needed as dependencies
  - preferred method -- be automatically loaded during reboot
  - can use `-v -n` options to preview what modprobe will do
    - -n: dry-run, This option does everything but actually insert or delete the modules
- Removing modules: `rmmod fat (File Allocation Table)`
  - will not remove module if currently in use, unless forced
  - Keep the kernel as lightweight as possible

Conclusion: modprobe is a more sophisticated tool compared to lsmod . It not only lists modules but also intelligently handles loading and unloading modules and their dependencies.

Windows Server

- Windows Server main release: NT, 2003, 2008, 2012, 2016, 2019, 2022, 2025
- Multiple versions of Windows Server exist.
- Each version is defined to meet the needs of a certain market segment

Versions Include:

  - Standard: Physical or minimally virtualized environments
  - DataCenter: Highly virtualized datacenters and cloud environments
  - Essential: Small businesses with up to 25 users and 50 devices

- Windows Server 2025
  - It was released on 01/11/2024, Standard, Datacenter, Essentials and Azure Datacenter

Roles and Features in WIN Server

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- You can dedicate an entire computer to one role or install multiple server roles on a single computer

- Each **role** has 1 or more services associated with it
- **Server Manager** is the tool that is used to install, configure, and remove Server Roles

- Features provide auxiliary or supporting functions to servers
- Typically, administrators add features to add functionality of installed roles, e.g., DHCP server on Windows Server

Typical roles

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- Active Directory Domain Services
- Active Directory Lightweight Directory Services
- DHCP Server
- DNS Server
- Web Server (IIS)
- File Services
- Print Server
- Streaming Media Services
- Windows Server Virtualization (Hyper-V)

You will configure those servers with red ink in the following labs

Disk partitions – manual/auto

- Why partition ? To build different sections in a building
  - Different filesystem types for different partitions
  - Better disk space management
  - Multi-OS support
- Auto partition
  - Typically, 3 partitions – 1) /boot (bootloader), 2) / (system), and 3) swap (RAM⇔H/D)
  - Root partition managed by LVM (Logical Volume Manager)
- Logical Volume Manager (LVM)
  - Combine multiple individual hard drives or disk partitions into a single volume group (VG). Subdivide VG into logical volume (LV)
  - Easy to add, remove, expand and shrink partitions

MBR (PC) partition scheme

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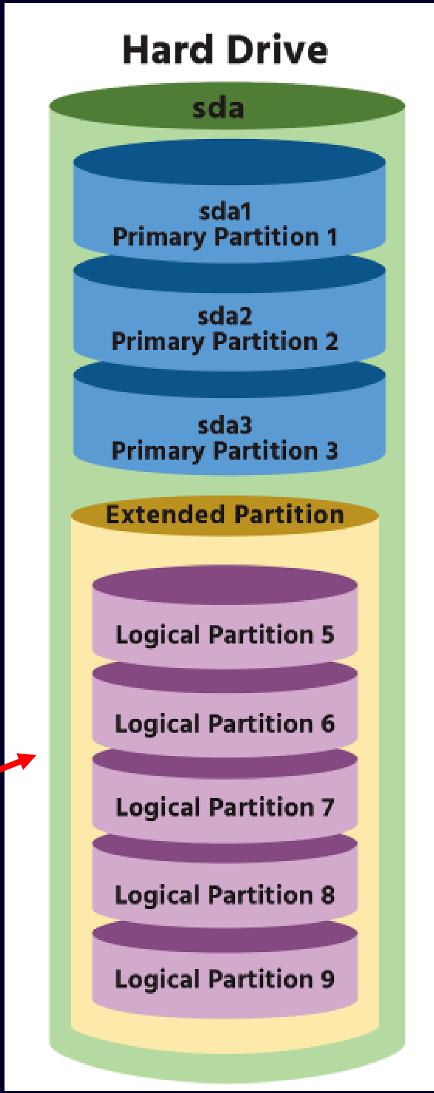
- Based on 1980's BIOS based systems (DOS & IBM-PC)
- Max 4 partitions (& max drive: 2TB!)
  - /dev/sda1, /dev/sda2, /dev/sda3, and /dev/sda4
- One of the 4 partitions, can allow an extended partition which allows "logical partitions" within it, starting from 5 (max 60), /dev/sda1-ada3, /dev/sda5
- P,P,P,P or P, P, P, E(L1,L2,…)

* MBR: Master Boot Record



GUID Partition Table

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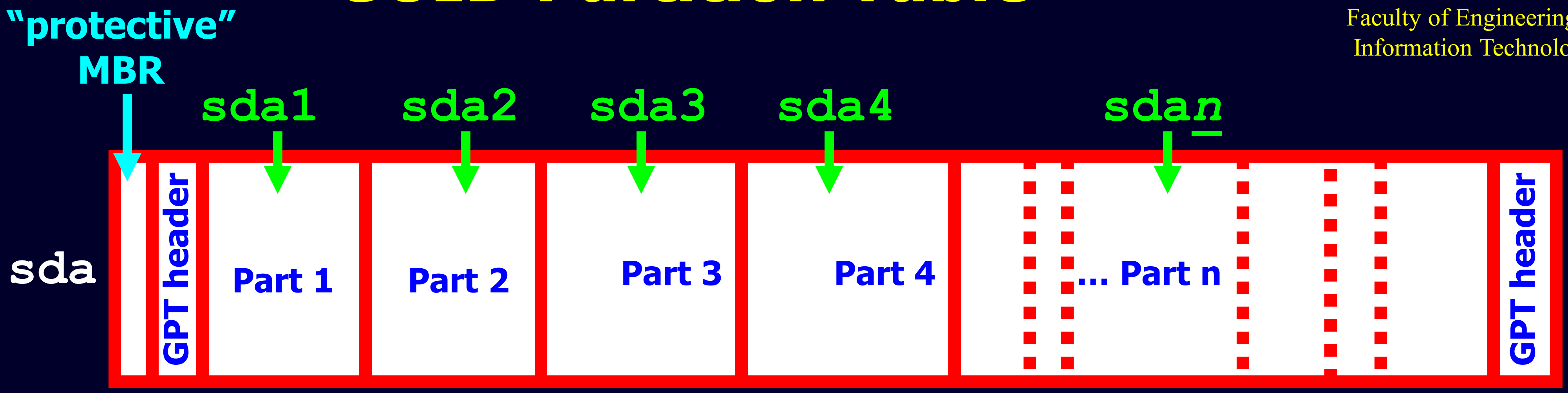

Modern configurations: UEFI based machines

- Replace MBR 4 partitions with 128: Max disk size: 9 ZB
  - 1M=1024K, 1G=1024M, 1T=1024G, 1P=1024T… K, M, G, T, P, E, Z, Y, B…
- Each partition now ID by a **GUID** (Globally Unique ID)
- Eg: Windows/Linux Basic Partition: {EBD0A0A2-B9E5-4433-87C0-68B6B72699C7}
- GPT (GUID Partition Table) header backed up at end of drive

Filesystems

- A file system is a data structure used for storing files on a disk partition
  - Directory structure indexing performance and file volume
  - Older Linux machines use ext2 (an extended file system type)
  - Formerly common: ext3 (ext2 + journal)
  - Ext4 (type 4)
    - ext4 is backward-compatible with ext3 and ext2
    - Support large file – up to 16 TB
    - Provide journaling function: tracking data for recover
  - btrfs: newer, high-performance, up to 16 EiB = 2^{64} bytes, support RAID
  - Alternatives file systems: ReiserFS, JFS, XFS (default for RHEL 7)
  - Compatible file systems: FAT32, exFAT
  - Non-Linux partitions:
    - e.g. NTFS, VFAT, HFS, ISO-9660, others
  - swap is technically not a file system, but usually treated in a similar way, create virtual memory using space on a physical device

Filesystems Hierarchy Standard (FHS)

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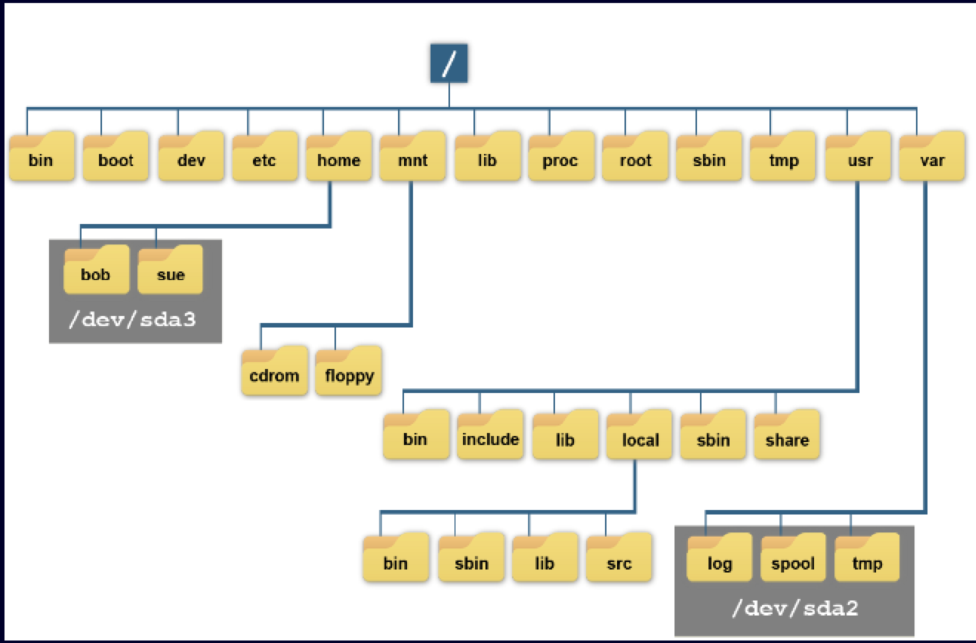
- In Linux system, users do not access files that are stored on partitions by directly using device file names(/dev/sda1,/dev/sda2..).
- The process of placing a filesystem under a mount point is called mounting.



Designing hard disk layout*

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Partition (mount point)	Typical size	Use
swap (not mounted)	1.5 – 2 times system RAM size	Adjunct to system RAM (virtual memory)
`/home`	200MB – 200GB	Users' directories: isolated for protection/upgrade
`/boot`	5MB – 50MB	System boot files, kernel, etc – for older BIOS's
`/usr`	500MB – 6GB	Unix system resources, including most programs and data files
`/usr/local`	100MB – 3GB	Programs/data unique to this installation
`/opt`	100MB – 3GB	Third-party software/data (often commercial)
`/var`	100MB – 200GB	Log files, mail/print queues, etc. – day-to-day use
`/tmp`	100MB – 20GB	Users' temporary files, sometimes *tmpfs*
`/mnt`	N/A	`/mnt` subdirectories are local/temp mount points
`/media`	N/A	Like `/mnt`, but for removable media (CD, USB, ...)

* some most common mount point

Creating partitions and filesystems

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- Create/edit partition table
  - `fdisk` command – create MBR partition only
    - each partition has a partition ID/type (really just a "hint" to the OS)
      - **0x83** = Linux data (can be ext2/ext3/ext4/reiserfs/jfs/xfs)
      - **0x82** = Linux swap
      - **0x07** = NTFS, 0x0C = Win95 FAT32, 0xAF = HFS, etc.
  - `gdisk` command – create GPT partition, support GPT indexing method
  - `parted` command – provide GUI interface, `gparted is` its GNU program,
- `mkfs` command – make filesystem (cf. formatting disk)
  - `mkfs -t ext4 /dev/sda3`
    - really just invokes `mkfs.ext4` command
  - Typical options include:
    - `-t` = type of filesystem
    - `-c` = bad block check
- `mkswap` command – make swap space
  - `mkswap /dev/sda4`
  - `swapon /dev/sda4`

Maintaining filesystem

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- Retrieve filesystem stats
  - **df**: Displays disk usage by partition
  - **du**: Displays disk usage by directory; good for finding users or applications that are taking up the most disk space
  - **iostat**: Displays a real-time chart of disk statistics by partition
  - **lsblk**: Displays current partition sizes and mount points

- Other ext2/ext3/ext4 filesystem commands:
  - **blkid** = display UUIDs and LABELs
  - **dumpe2fs -h** = display detailed filesystem info
  - **tune2fs** = adjust tuneable parameters
  - **debugfs** = interactively debug and repair damage (advanced)

Mounting filesystems

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- `/etc/fstab` is filesystem table (what can be mounted)
  - The filesystems listed in /etc/fstab gets mounted during booting process.
  - `/etc/fstab` is mount table (what is currently mounted)
- Example fstab entries:

```
# Device     Mount point          FS type      FS options      F P
/dev/sda2    /                    ext3         defaults        1 1
/dev/sda1    /boot                ext3         defaults        1 2
/dev/sda5    /home                ext3         defaults        1 2
/dev/sda3    swap                 swap         defaults        0 0
/dev/sdb     /media/cdrom         iso9660      defaults        0 0
```

F = backup using dump utility (1 = yes, out of date), P = order of FS checking at boot (/ root first)
0 means ignore

- Mounting:
  - `mount /media/cdrom` (if in fstab)
  - `mount -t vfat /dev/sdb1 /media/usbdisk` (not in fstab)
- Unmounting:
  - `umount /media/cdrom` (works regardless of fstab)

Managing software using packages

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- Modern operating systems typically manage software as packages
  - Package includes executables, config files, documentation, etc.
  - **Package contents spread across many directories on the filesystem**
  - For Linux, two main package managers:

- RedHat (RPM Package Manager, RPM) – .rpm files:
  - Red Hat, Fedora, CentOS, openSUSE, OpenMandriva Lx
    - **rpm [operation] [options] [package-files|package-names]**
      - `-i` = install, `-U` = upgrade or install, `-q` = query
    - **Name of the software package**: ***`packagename-a.b.c-x.arch.rpm`***
      a.b.c = version, *x* = build/release no., *arch* = hardware, *e.g. x86_64.rpm (extension)*

    Ex: rpm -i packagename-1.0.0-1.x86_64.rpm

- Debian (Apt) – .deb files: Ubuntu, dpkg is the package manager
  - Naming: ***`packagename_a.b.c-x_arch.deb`***
    - similar to RPM, some different arch's (powerpc, all, ...)
    - `dpkg` *`[options] [action] [package-files|package-name]`*
    - `-i` = install, `-r` = remove package

Yum/Dnf – Red Hat

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YUM is a front-end utility that uses the RPM package manager for package management

- `yum` = Yellowdog Updater, Modified
  - tool for keeping a system's RPM packages up to date
  - **Using repository to solve dependency problem**
  - YUM repository configuration files are stored in `/etc/yum.repos.d/*`
- How to use:

```
yum install mysql      - install mysql package
yum update             - update all installed packages
yum check-update       - see if updates are available
yum upgrade            - like update, but handles obsolete pkgs
yum remove mysql       - removes a package (also yum erase)
yum list updates       - list available updates (more options)
yum search mysql       - search descriptions for mysql
yum clean packages     - cleans up /var/cache/yum/...
```

apt-get - Ubuntu

- `apt-get` = Advanced Packaging Tool "get" utility
  - tool for keeping system's Debian packages up to date
  - Similar to yum for RPM
  - repositories configured in /etc/apt/sources.list
- e.g.

`apt-get install mysql`	- install mysql package
`apt-get update`	- list available updates
`apt-get upgrade`	- upgrade all installed packages
`apt-get dist-upgrade`	- upgrade with conflict resolution
`apt-get remove mysql`	- removes a package
`apt-get clean`	- cleans up `/var/cache/apt/...`
`apt-get autoclean`	- cleans no-longer-available pkgs

Viewing processes & Job control

- View running processes with:
  - **ps**: Support Unix-style option (-), BSD-style (none), and GNU style (--)

`ps`	- view processes on current terminal
`ps -ef`	- "e" = all processes, "f" = full format
`ps -u root --forest`	- "u" specifies user, "forest" = hierarchy
`top`	- dynamic view of processes

  - `top`: show real-time information
    - `free -h` show memory usage
- Job control

`jobs`	- show jobs in current session
`somecommand &`	- run "somecommand" in background
Ctrl-C	- Ctrl-C will **permanently kill** current fg job
Ctrl-Z	- Ctrl-Z will **temporarily stop** current fg job
`fg 2`	- bring job #2 to foreground
`bg 2`	- send job #2 to background

Killing Processes & Sending Signals

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- Killing processes
  `kill 12345` - send SIGTERM to Process ID 12345
- Sending other signals
  `kill -9 12345` - sends signal 9 (SIGKILL) to PID 12345
- SIGKILL (9) – Kill signal. Use SIGKILL to forcefully terminate a process. This will not save data or cleaning kill the process.
- SIGTERM (15) – Termination signal. This is **the default and safest way** to kill process.

```
[root@localhost ~]# kill -l
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL       5) SIGTRAP
 6) SIGABRT      7) SIGBUS       8) SIGFPE       9) SIGKILL     10) SIGUSR1
11) SIGSEGV     12) SIGUSR2     13) SIGPIPE     14) SIGALRM     15) SIGTERM
16) SIGSTKFLT   17) SIGCHLD     18) SIGCONT     19) SIGSTOP     20) SIGTSTP
21) SIGTTIN     22) SIGTTOU     23) SIGURG      24) SIGXCPU     25) SIGXFSZ
26) SIGVTALRM   27) SIGPROF     28) SIGWINCH    29) SIGIO       30) SIGPWR
31) SIGSYS      34) SIGRTMIN    35) SIGRTMIN+1  36) SIGRTMIN+2  37) SIGRTMIN+3
38) SIGRTMIN+4  39) SIGRTMIN+5  40) SIGRTMIN+6  41) SIGRTMIN+7  42) SIGRTMIN+8
43) SIGRTMIN+9  44) SIGRTMIN+10 45) SIGRTMIN+11 46) SIGRTMIN+12 47) SIGRTMIN+13
48) SIGRTMIN+14 49) SIGRTMIN+15 50) SIGRTMAX-14 51) SIGRTMAX-13 52) SIGRTMAX-12
53) SIGRTMAX-11 54) SIGRTMAX-10 55) SIGRTMAX-9  56) SIGRTMAX-8  57) SIGRTMAX-7
58) SIGRTMAX-6  59) SIGRTMAX-5  60) SIGRTMAX-4  61) SIGRTMAX-3  62) SIGRTMAX-2
63) SIGRTMAX-1  64) SIGRTMAX
```

Process Priorities

- Priorities
  - Numeric value associated with a process
  - Used by OS (scheduler) to apportion CPU time to processes
  - High priority processes likely to get more CPU time
  - Different Unixes differ in numeric ranges and algorithms used to implement priorities
  - Common to all Unixes and Linux is the "nice" value

- Nice value
  - In early Unix, nice users running long jobs would be nice to other interactive users
  - A "nice" process lets other processes go first, and has a high nice value
  - High nice value means low priority (lets others go first)

-20	0	19
Highest priority	Default niceness	Lowest priority

Nice Values

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- Nice values range from -20 to +19
  - negative nice values have highest priority
  - default nice value for newly created processes is 0
  - only root can increase a process priority (decrease niceness)

- Creating new processes with defined priorities

  `nice myprog` - no nice value, defaults value is 0
  `nice +12 myprog` - runs with nice +12 (or **`nice –n 12 myprog`**)
  `nice –n –12 myprog` - runs with nice –12

- Change priority of running process

  `renice -20 12345` - set PID 12345 to nice -20 (max)
  `renice 19 –u chen anup` - nice 19 for chen & anup's processes

Windows

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- Use task manager (Ctrl-Alt-Del)
  - Choose processes tab or Details tab for even more
  - You can view/sort processes by clicking header
    - If admin authority, also click Show Process from All users
  - **Delete/Change priority by Right clicking process & choosing action**
  - Windows priorities:
    - Realtime, high, above normal, normal, below normal, low
    - You can also set affinity i.e.: which CPU the process can use

- Command line equivalents on PowerShell

  `tasklist` - list tasks, lots of filtering options

  `taskkill` - kill tasks

Q&A

- What raw device file would Linux create for the second SCSI drive connected to the system?
  A. /dev/hdb B. /dev/sdb C. /dev/sdb1 D. /dev/hdb1 E. /dev/sda

- Scott wants to add a third-party repository to his Red Hat–based package management system. Where should he place a new configuration file to add it?
  A. /etc/yum.repos.d/ **B.** /etc/apt/sources.list **C.** /usr/lib/
  D. /bin/ **E.** /etc/

- Scott has decided to run a program in the background due to its time to process. However, he realizes several hours later that the program is not operating correctly and may have been consuming large amounts of CPU time unnecessarily. He decides to stop the background job. What command should he first employ?
  A. Scott should issue the ps -ef command to see all his background jobs.
  B. Scott should issue the jobs -l command to see all his background jobs.
  C. Scott should issue the kill %1 command to stop his background job.
  D. Scott should issue the kill 1 command to stop his background job.
  E. Scott should issue the kill -9 1 command to stop his background job.